

A Post-processing Ensemble Machine Learning Approach for Prediction and Classification of Cardiovascular Disease

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Abstract—Cardiovascular Disease (CVD) is one of the dangerous diseases which affecting the human health. CVD is the significant cause of mortality as well as morbidity in the population of the world. The prediction and diagnosis of CVD is challenging in clinical data analysis as well as health care providers to avert the patient from diminishing like disease as well as preserve lives. An automatic detection for the classification of arrhythmia performs a significant role in managing as well as curing the cardiovascular diseases. Therefore, this research proposes the post-processing ensemble Machine Learning (ML) approach for the prediction and classification of CVD. Initially, this research collected the publicly available Kaggle dataset of CVD for validate the effectiveness of the model. Then, handling missing values as well as data unbalancing techniques are used for the data pre-processing. After that, Principal Component Analysis (PCA) is used for the selection of relevant features. Finally, post-processing ensemble ML approach is used for the classification of CVD. The proposed method is estimated by various performance metrics and it achieves the accuracy of 99.56% when compared to the previous approaches such as Support Vector Machine (SVM) and Gradient Boosting (GB).

Keywords—cardiovascular disease, data imbalance, missing values, post-processing ensemble machine learning and principal component analysis.

I. INTRODUCTION

According to the World Health Organization (WHO) discovered that the cardiovascular disease (CVD) is measuring the leading cause of the death [1]. The CVD is the significant cause of death as well as death rate is increased every year. The CVD is caused through artery damage to the entire part as well as insufficient nutrient as well as oxygen supply to the heart. There are various types CDS like coronary heart disease, hypertensive heart disease, stroke rheumatic as well as inflammatory and so on [2]. Subsequently, various researches have been performed to execute in computer-assisted diagnosis systems. Nevertheless, some approaches have leads to suitable outcomes in real-time conditions [3]. Thus, number of researchers have been taken out to develop the various number of diagnosis and prediction approaches of the CVD. The manual prediction is complex because of its redundancy, incompleteness as well as near suggestion by

time in medical data [4]. To overcome these challenges, the researchers introduced the Machine Learning (ML)-based approaches for the prediction of CVD as well as it has the capability to determine the large number of difficult data as well as provide the better accuracy results [5]. The novelty of this research is contributed in the following:

- This research proposes the post-processing ensemble ML approach which involves Support Vector Machine (SVM), Random Forest (RF) as well as Gradient Boosting (GB) for the prediction and classification of CVD.
- This research utilizes the pre-processing technique of handling missing values as well as handling data imbalance to enhance the input data and to enhance the accuracy.
- The Principal Component Analysis (PCA) is utilized for the feature extraction process to efficiently select the relevant features from the pre-processed data.

The rest of the paper is arranged as following: Section 2 discusses the literature survey. Section 3 provides the proposed methodology. Section 4 presents the experimental results and Section 5 gives the conclusion of overall paper.

II. LITERATURE SURVEY

Mohammed Nasir Uddin and Rajib Kumar Halder [6] presented the ensemble approach-based multi-layer dynamic system (MLDS) for the prediction of cardiovascular disease. The presented approach utilized the Information Gain Attribute Evaluator (IGAE), Correlation Attribute Evaluator (CAE), Gain Ratio Attribute Evaluator (GRAE), Extra Trees classifier (ETC) as well as Lasso for the selection of features. Eventually, the ensemble approach involves Random Forest (RF), Gradient Boosting (GB), Naïve Bayes (NB) for the classification. However, the presented approach had maximum time complexity during training.

Manoj D. Patil and Dr. Harsh Mathur [7] developed the Apache Spark-based real-time prediction system according to the numerous distributed computing process. In that approach, an Advanced Absolute Shrinkage and Selection Operator

(AASSO) function was utilized for the efficient feature selection. The developed approach contained two significant types such as streaming as well as data storage and display. The initial approach combined the Spark MLlib by streaming as well as then performed for the classification. However, the developed approach of AASSO had even selected the greater number of features, K-Nearest Neighbor (KNN) developed poor results.

David Opeoluwa Oyewola et al. [8] introduced the ensemble Deep Learning (DL) approach for the prediction CVD according to the complex factors. The effectiveness of every approach was estimated with various approaches like Long Short-Term Memory (LSTM), Feedforward, Cascade Forward Neural Network (CFNN) as well as Elmann Neural Network (ELN). The suggested approach efficiently classifies all the input data and it achieved better accuracy. However, the introduced approach had required the greater number of data to train the model.

Aqsa Rahim et al. [9] implemented the ML-based CVD diagnosis called MaLcADD for an efficient CVD prediction with maximum precision. Specifically, the suggested approach initially committed with the missing values as well as data imbalance by the utilization of mean replacement approach as well as Synthetic Minority Over-sampling Technique (SMOTE). Feature significance approach was used

for the selection of features as well as ensemble approach of Logistic Regression (LR) as well as KNN was used for the prediction and classification of CVD. However, the suggested approach had utilized only limited datasets to train the model.

Hana H. Alalawi and Manal S. Alsuwat [10] developed to perform the various classification approaches such as Support Vector Machine (SVM), LR, KNN, NB, and DT for the classification of CVD. Furthermore, developed the deep network as well as calculated the influence of different optimization learning approaches for the detection of CVD based on the identification of appropriate classifier. The residual gradient final time was minimized in every estimation but, lack of intelligent approach in selection of subset features.

III. PROPOSED METHODOLOGY

In this research, the post-processing ensemble ML approach is proposed for the efficient prediction and classification of CVD. This research comprises four significant steps: dataset collection using Kaggle CVD dataset, pre-processing by handling missing values as well as handling data imbalance techniques, feature extraction using PCA and classification is done by using post-processing ensemble ML approach. Figure 1 depicts the workflow of the suggested method.

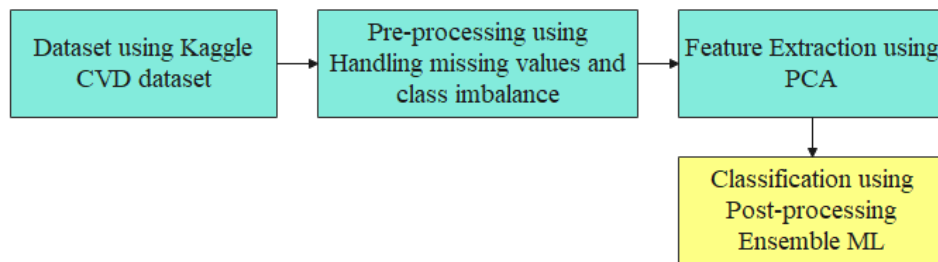


Fig. 1. Workflow of the suggested method

A. Dataset

The initial process of this research is the collection of datasets to validate the performance of the proposed method. This research collected the benchmark dataset of CVD dataset utilized for the prediction and diagnosis process. This acquired dataset consists of 70000 records with 11 autonomous features which collected from Kaggle. The data are acquired at the instant of medical determination as well as data provided through the patient's data and their medical recordings. The collected dataset is divided into two parts such as 80% of the data is used for the training and remaining for the testing process. The collected dataset is then provided as input to the pre-processing of the data.

B. Pre-processing

In this section, the pre-processing utilizes the input as the collected dataset. In this step, initially, the input data is identified for the possible outliers and the by the utilization of boxplots, the outliers of the input data are removed. Generally, the collected dataset contains missing values and which leads to the poor performance. The mean substitution is utilized for the handling the missing values through the mean of every value of the corresponding attribute. After that, the SMOTE technique is used for handling the class imbalance problem. SMOTE is utilized for enhancing the samples for the minority classes from majority classes as well as for data augmentation

surveyed the KNN approach which developed the non-overlapping synthetic samples for minority classes. The enhanced samples through this approach are similar to the actual data which enhances the flexibility as well as minimizes the randomness. Then, the outcome of the handled class imbalance data is provided as input to the feature selection process.

C. Feature Selection

The output of pre-processed data is provided as an input for the process of feature selection and it also known as selection of attribute. The features selection reduces the training time as well as eliminate the computational complexity. This process can enhance the model accuracy due to the correct subset selection. In this feature selection process, the Principal Component Analysis (PCA) approach is used for the selection of relevant features. PCA is a mathematical approach for the minimization of data dimensionality through determining as well conserving the data with describing the directions. PCA is an extraction of dominant patterns in a matrix of complementary score sets from linear algebra as well as statistics to estimate the dimensionality. It seems like sensitive to an outlier as well as efficient grouping in a plot. Them selected features are fed to the classification process.

D. Classification

After feature selection, the post-processing ensemble ML-based techniques are used for prediction and classification of CVD. The selected features output is further proceeded to the classification of images in CVD. Here are some commonly used ensembles ML algorithms for this task. This section develops the postprocessing approaches which are methodologically compared for ensemble ML classification of the CVD. The postprocessing approach have introduced in multiple sets, begin with accomplished. This research has ensued to ML approaches ranging from RF to Gradient Boosting approaches based on the Neural Network. Considered that for every post-processing approach, this research sets the individual approach for every lead time according to the training data.

1) *Support Vector Machine*: The SVM is significantly utilized for the problem of classification and it develops the hyperplane as dividing the data into number of classes. SVM have the capability to address the problem of both linear as well as non-linear. The goal of SVM is to identify the hyperplane in N-dimensional space. Which efficiently classifies the datapoints. Initially, SVM maps the non-linear training sample with the high-dimensional space, hence, the linear non-dividable data below less-dimensional situations are integrated to the high-dimensional space for designing it linearly separable data. As an outcome, the hyperplane is efficiently classifying the datapoint as well as maximal interval is addressed.

2) *Random Forest*: Random Forest is developed based on the number of Decision Trees (DT), and the result of this approach is identified through the different tree result. Particularly, DT choose the optimal features from present feature sets. Nevertheless, RF has arbitrarily chosen the subsets from a node set which depends on the basic DT, after it selects the optimal features from the subsets. RF has greater performance and which is utilized for large data set as well as it has the capability to provide better performance result for high-dimensional data.

3) *Gradient Boosting*: GB is one of the most ML approach, which is utilized for the classification problem. This approach develops the system in a stage-wise fashion, after simplifies the system by permitting the optimization of a random variable loss function, where every classifier simplifies the error of the precursor. This approach is depending on the development of DT. Primarily, DT is developed as well as trained on the utilization of the dataset, after it has validated according to another DT is developed as well as in which enhancement is develop. This procedure is continued for the particular number of iterations. Subsequently, the last system is the weighted sum of the classifiers developed over the previous tree approaches.

IV. EXPERIMENTAL RESULTS

The proposed method for CVD is executed by utilizing the platform of python 3.8 with Windows 10 OS, 16GB RAM with intel-i7 processor. The proposed method is analyzed based on various assessment metrics like accuracy, precision, recall and F1-score. The mathematical expressions of these metrics are expressed in (1)-(4) as:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$Precision = \frac{TN}{TN+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$F1 - score = \frac{2TP}{2TP+FP+FN} \quad (4)$$

Where, TP - True Positive, TN - True Negative, FP - False positive, and FN - False Negative.

A. Performance Analysis

In this section, the performance analysis of the proposed post-processing Ensemble ML is estimated with the Kaggle CVD dataset. Table 1 and 2 shows the performance analysis of feature extraction and classification results.

TABLE I. PERFORMANCE ANALYSIS OF FEATURE EXTRACTION RESULTS

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LDA	88.56	91.26	81.35	82.69
MI	91.48	92.19	83.37	84.36
PCC	94.35	95.38	85.37	86.36
Chi-square	97.67	96.49	87.46	89.67
PCA	99.56	99.02	89.34	91.25

As shown in Table 1, the performance analysis of feature extraction results of the proposed method with previous works are discussed. The effectiveness of the proposed method is validated by utilizing the various performance metrics on the Kaggle CVD dataset. The effectiveness of the PCA is analyzed as well as validated by using the previous feature extraction approaches like Linear Discriminative Analysis (LDA), Mutual Information (MI), Pearson Correlation Coefficient (PCC) and Chi-square. The proposed PCA approach achieves the accuracy of 99.56%, precision of 99.02%, recall of 89.34% and F1-score of 91.25 respectively. As compared the results of these existing feature extraction approaches, the proposed PCA achieves better results in overall metrics.

TABLE II. PERFORMANCE ANALYSIS OF CLASSIFICATION RESULTS

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LR	91.35	92.58	83.38	85.38
SVM	93.36	94.38	85.38	87.33
RF	95.47	96.37	86.27	89.36
GB	98.46	97.47	88.34	90.26
Post-processing Ensemble ML	99.56	99.02	89.34	91.25

As shown in Table 2, the performance analysis of classification results of the proposed method with previous works are discussed. The effectiveness of the proposed method is validated by utilizing the various performance metrics on the Kaggle CVD dataset. The effectiveness of the Post-processing Ensemble ML is analyzed as well as validated by using the previous classification approaches like LR, SVM, RF as well as GB. The proposed Post-processing Ensemble ML approach achieves the accuracy of 99.56%, precision of

99.02%, recall of 89.34% and F1-score of 91.25 respectively. As compared the results of these existing feature extraction approaches, the proposed Post-processing Ensemble ML achieves better results in overall metrics.

B. Comparative Analysis

Table 3 shows the comparison results of the proposed approach with the previous works using CVD dataset. The effectiveness of the proposed is analyzed and compared with the previous works based on the various performance metrics like accuracy, precision, recall and F1-score.

TABLE III. COMPARISON RESULTS OF THE PROPOSED METHOD USING KAGGLE CVD DATASET

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
MLDS [6]	94.16	97	N/A	N/A
AASSO [7]	92.85	98	N/A	N/A
ELMAN [8]	72.74	N/A	N/A	N/A
MaLCaDD [9]	99.1	N/A	N/A	N/A
Ensemble -GB [10]	70	70	70	70
Proposed post-processing Ensemble ML	99.56	99.02	89.34	91.25

C. Discussion

In this section, the benefits of the proposed method as well as drawbacks of the previous works are discussed. The drawbacks such as MLDS-GB [6] had maximum time complexity during training. AASSO [7] had even selected the greater number of features, K-Nearest Neighbor (KNN) developed poor results. ELN [8] required the greater number of data to train the model. MaLCaDD [9] utilized only limited datasets to train the model. Ensemble approach [10] had lack of intelligent approach in selection of subset features. The proposed method has solved these challenges by performing the ensemble approach-based prediction and classification of CVD based on the complex factors. The proposed method has the advantage of minimizing the variance in the approaches, thus enhancing the performance. The proposed method achieves the accuracy and precision of 99.56% and 99.02% respectively when compared to the previous approaches.

V. CONCLUSION

Globally, CVD is the most predominant disease and which leads to death. In this research, the efficient post-processing Ensemble ML approach is proposed for the prediction and classification of the CVD. The post-processing Ensemble ML approach involves three main approaches such as SVM, RF as well as GB. In this research, the Kaggle CVD dataset is collected to estimate the effectiveness of the model. The experimental results shows that the proposed method achieves better results by utilizing the performing metrics such as accuracy, precision, recall and F1-score and it achieves the values of about 99.56%, 99.02%, 89.34%, and 91.25% respectively. In future, the proposed method will expand to utilize the Deep Learning (DL) approach to enhance the overall performance.

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